



Applied ML

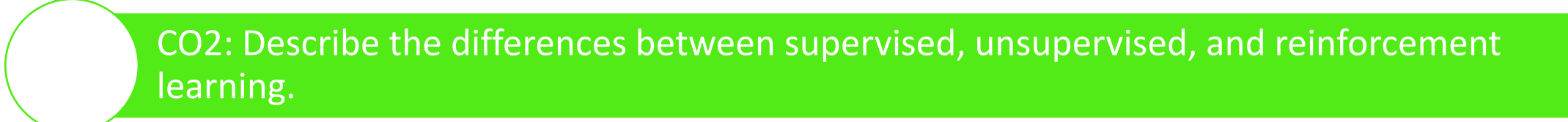
Prerequisite(s): Elements of AI

Total Units to be Covered: 6	Total Contact Hours: 48
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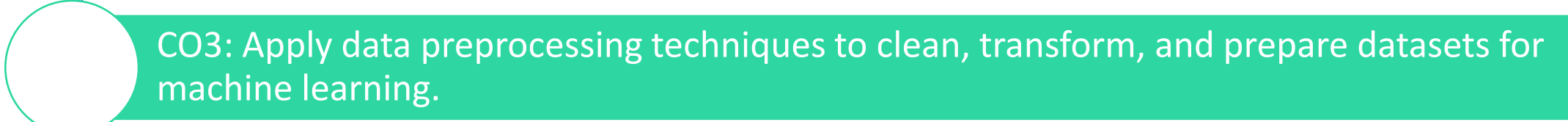
Course Outcomes



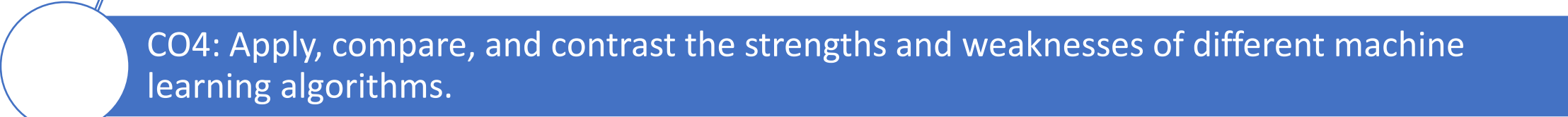
CO1: Recall and define key machine learning concepts, terminologies, and algorithms.



CO2: Describe the differences between supervised, unsupervised, and reinforcement learning.



CO3: Apply data preprocessing techniques to clean, transform, and prepare datasets for machine learning.



CO4: Apply, compare, and contrast the strengths and weaknesses of different machine learning algorithms.

Unit V: Regression

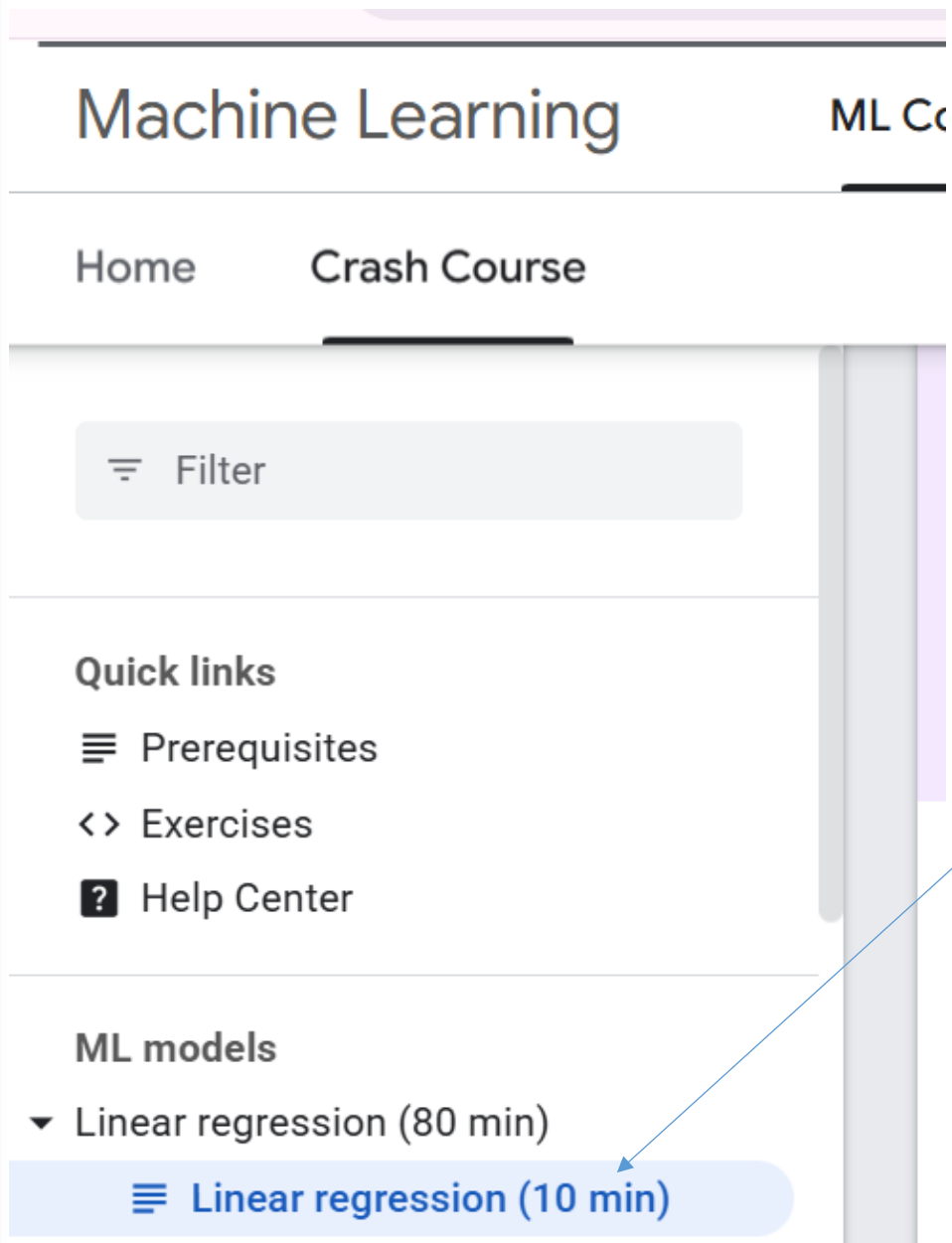
12 Lecture Hours

Introduction to Regression, Regression examples, Regression models, Steps in regression analysis, Linear regression, Simple linear regression, Mathematical proof of linear regression, Least squares estimation, Least squares regression-Line of best fit, Illustration, Direct regression method, Maximum likelihood estimation, Coefficient of determination (R-squared), Checking model adequacy, Over-fitting, Detecting over-fit models: Cross validation, Logistic regression, Mathematical proof of logistic regression, multiple linear regression, Multiple linear regression model building, Mathematical proof of Multiple linear regression model, Interpretation of multiple linear regression coefficients-Partial regression coefficients, Standardized regression coefficients, Missing data, Validation of multiple regression model, regularization, ridge and lasso regularization.

- Regression is a supervised learning technique which helps in finding the correlation between variables and enables us to predict the continuous output variable based on the one or more predictor variables.
- It is mainly used for **prediction, forecasting, time series modeling, and determining the causal-effect relationship between variables.**
- In Regression, we plot a graph between the variables which best fits the given datapoints, using this plot, the machine learning model can make predictions about the data.

"Regression shows a line or curve that passes through all the datapoints on target-predictor graph in such a way that the vertical distance between the datapoints and the regression line is minimum."

- The distance between datapoints and line tells whether a model has captured a strong relationship or not.
- Some examples of regression can be as:
 - Prediction of rain using temperature and other factors
 - Determining Market trends
 - Prediction of road accidents due to rash driving.



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Complete the LR 10 mins reading

ML Concepts

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ML models

▾ Linear regression (80 min)

 Linear regression (10 min)

 **Loss (10 min)**

 Interactive exercise:
Parameters (5 min)

 Gradient descent (10 min)

 Hyperparameters (10 min)

 Interactive exercise: Gradient
descent (5 min)

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Was this helpful?  

Linear regression: Loss



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On this page

Distance of loss

Types of loss

Calculating loss example

Choosing a loss

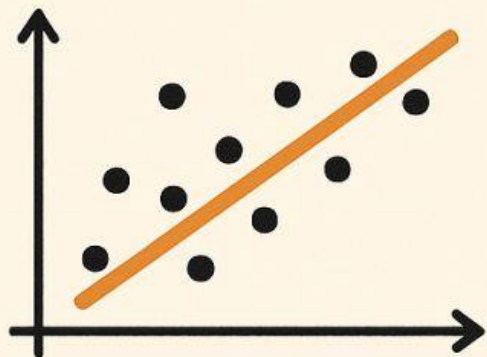
Check Your Understanding

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Crash Course.**

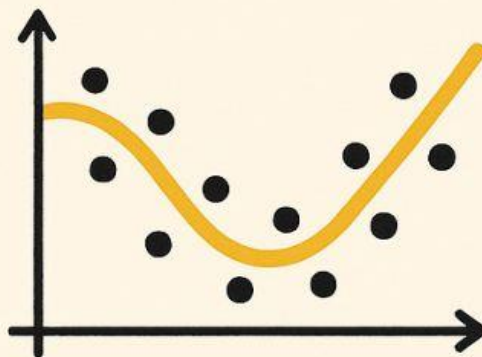
Complete the LR 10 mins reading

TYPES OF REGRESSION METHODS

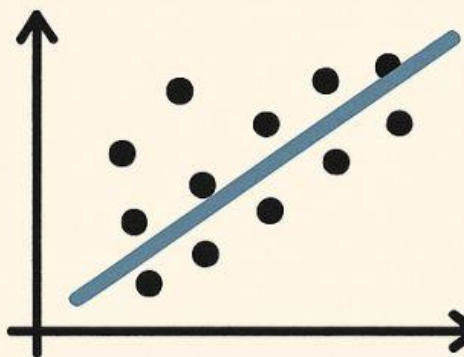
LINEAR REGRESSION



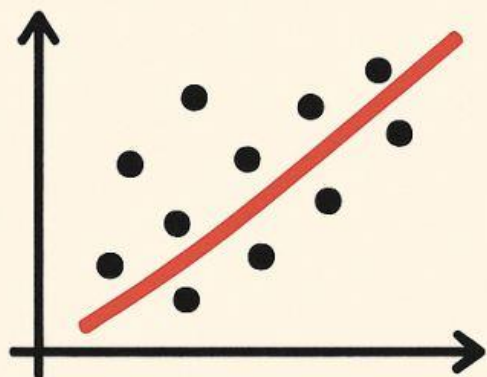
POLYNOMIAL REGRESSION



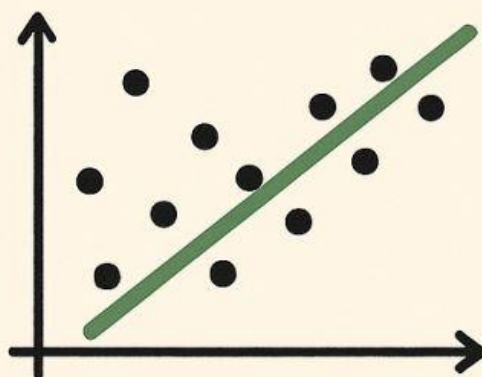
RIDGE REGRESSION



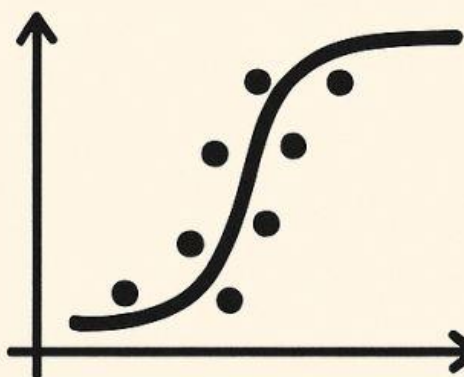
LASSO REGRESSION



ELASTIC NET REGRESSION



LOGISTIC REGRESSION



Types of Regression Methods

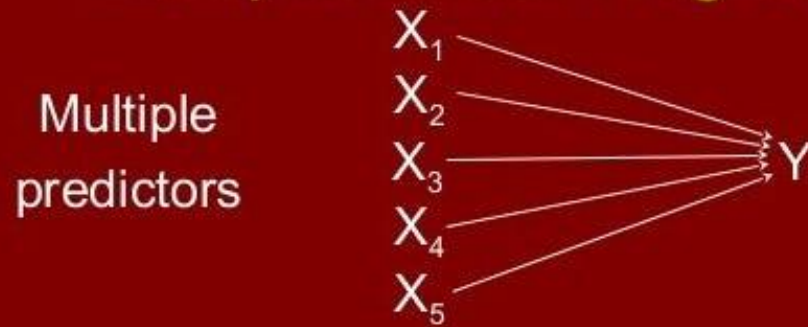
Regression analysis is a cornerstone of statistical modeling, offering various techniques to meet specific analytical needs. Here's a look at some essential regression methods:

- 1. Linear Regression:** This traditional technique establishes a straightforward linear relationship between one or more independent variables and a dependent variable.
- 2. Polynomial Regression:** Building on the foundation of linear regression, this method fits a polynomial function to the data, allowing it to accommodate more complex, non-linear relationships.
- 3. Ridge Regression:** As a robust variation of linear regression, ridge regression introduces a penalty term to the loss function, discouraging significant coefficients that can lead to overfitting.
- 4. Lasso Regression:** Similar to ridge regression, lasso regression applies a different type of penalty that helps prevent overfitting and performs automatic feature selection, making some coefficients zero.
- 5. Elastic Net Regression:** This versatile method blends ridge and lasso regression techniques. It combines the advantages of both approaches while still performing feature selection.
- 6. Logistic Regression:** While its name includes "regression," this technique is used for binary classification. It models the probability that a given input belongs to a particular category.

Linear Regression

Single predictor $X \longrightarrow Y$

Multiple Linear Regression



Regressions

Simple
Linear
Regression

$$y = b_0 + b_1x_1$$

Multiple
Linear
Regression

$$y = b_0 + b_1x_1 + b_2x_2 + \dots + b_nx_n$$

Polynomial
Linear
Regression

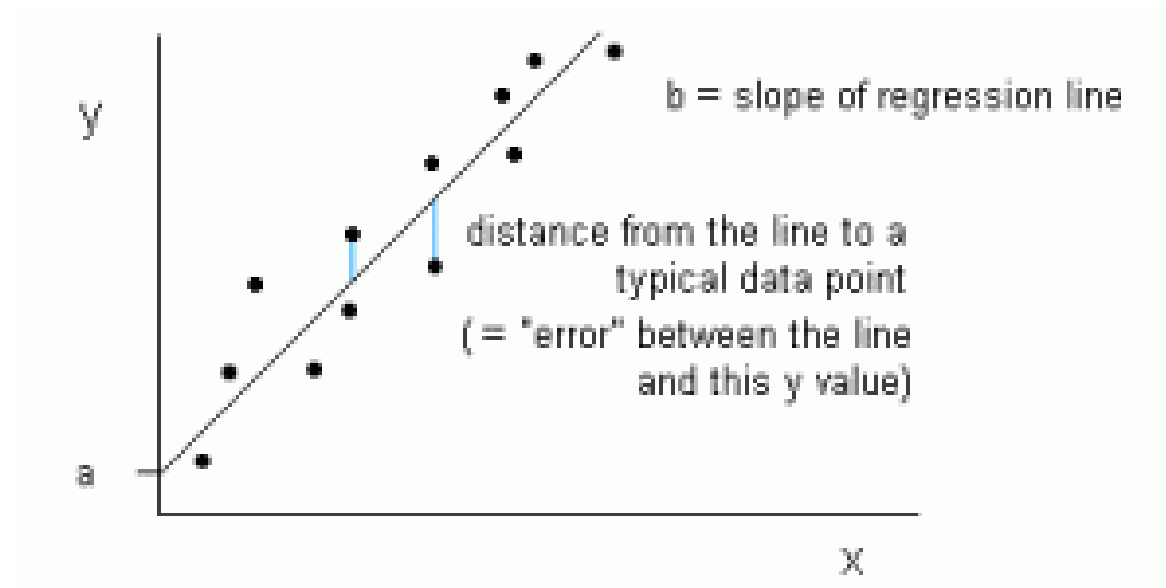
$$y = b_0 + b_1x_1 + b_2x_1^2 + \dots + b_nx_1^n$$

Linear Regression

- **Linear Regression model can be represented using the below equation:**

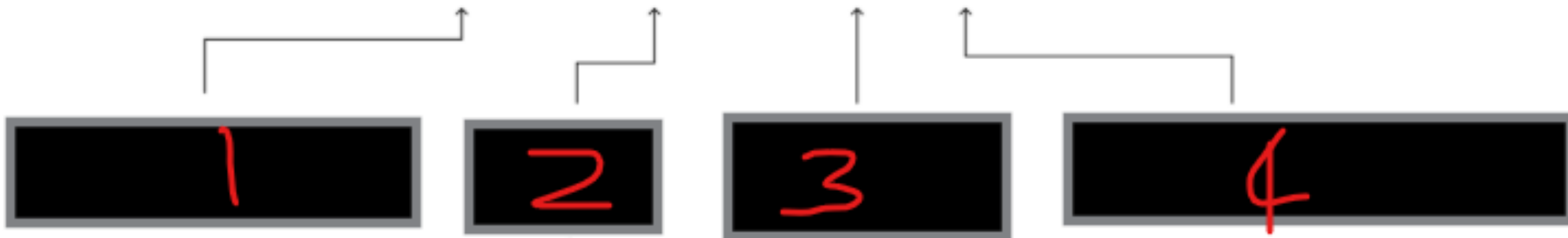
$$Y = a_1 + a_2 X + \epsilon$$

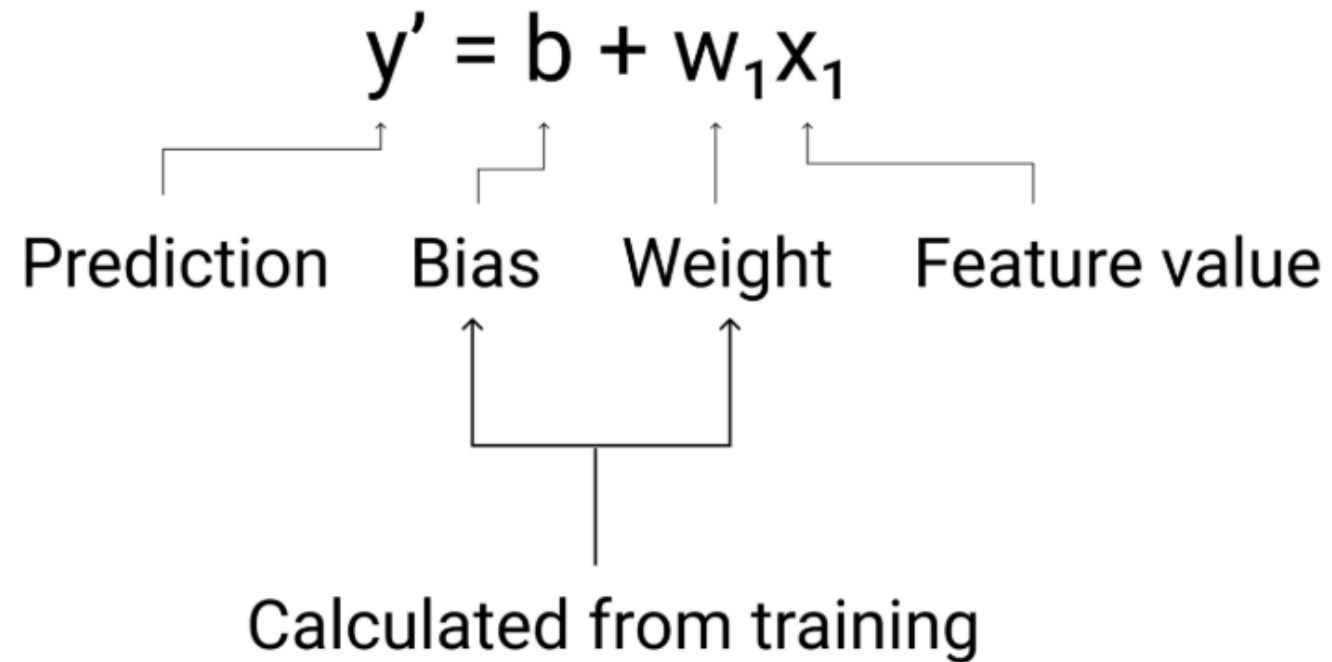
- Here,
- a_1 = It is the intercept of the Regression line (can be obtained putting $x=0$)
- a_2 = It is the slope of the regression line, which tells whether the line is increasing or decreasing.
- ϵ = The error term. (For a good model it will be negligible)



- Linear regression is a statistical regression method which is used for predictive analysis.
- It is one of the very simple and easy algorithms which works on regression and shows the relationship between the continuous variables.
- It is used for solving the regression problem in machine learning.
- Linear regression shows the linear relationship between the independent variable (X-axis) and the dependent variable (Y-axis), hence called linear regression.
- **If there is only one input variable (x), then such linear regression is called simple linear regression. And if there is more than one input variable, then such linear regression is called multiple linear regression.**
- The relationship between variables in the linear regression model can be explained using the below image. Here we are predicting the salary of an employee on the basis of **the year of experience**.

$$y' = b + w_1 x_1$$





Mathematical Derivation of Linear Regression

HOW TO DERIVE FORMULAS FOR SIMPLE LINEAR REGRESSION



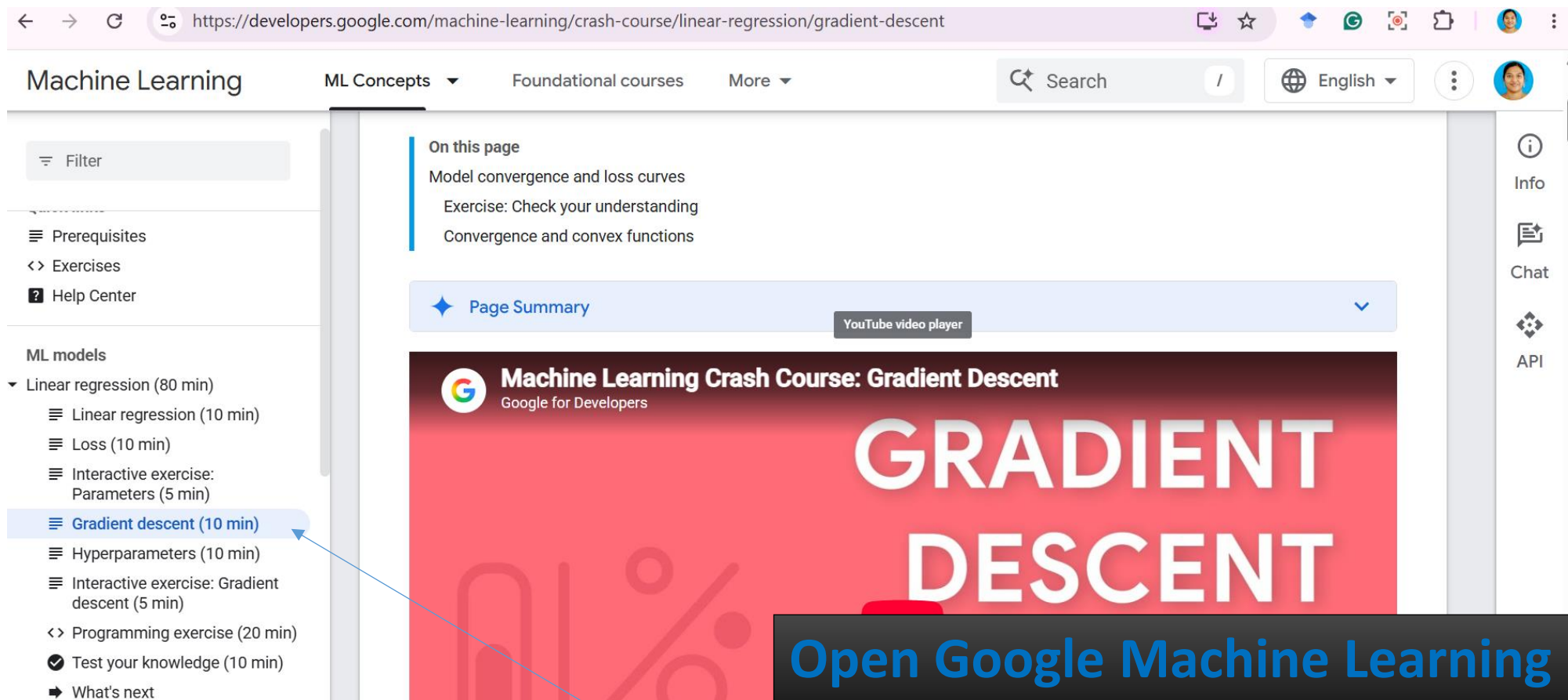
$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

$$\hat{\beta}_{0,OLS} = \bar{Y} - \hat{\beta}_1 \bar{X}$$

$$\hat{\beta}_{1,OLS} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

Aspect	OLS - Ordinary Least Squares	Gradient Descent
Basic idea	Finds coefficients mathematically	Finds coefficients iteratively
Approach	Direct / closed-form	Iterative / optimization
Uses	Least-squares equations	Loss function + gradients
Starting values needed?	✗ No	✓ Yes
Learning rate needed?	✗ No	✓ Yes
Iterations?	✗ No	✓ Yes
Main calculation	Solve for (β_0, β_1)	Update (β_0, β_1) repeatedly
Dataset size	Can become expensive for very large datasets	Good for very large datasets
Result	Exact OLS solution	Approaches the minimum
Common in	Statistics, classical regression	Machine Learning



Machine Learning

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Prerequisites

Exercises

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ML models

- Linear regression (80 min)
 - Linear regression (10 min)
 - Loss (10 min)
 - Interactive exercise: Parameters (5 min)
 - Gradient descent (10 min)**
 - Hyperparameters (10 min)
 - Interactive exercise: Gradient descent (5 min)
- Programming exercise (20 min)
- Test your knowledge (10 min)
- What's next

On this page

- Model convergence and loss curves
- Exercise: Check your understanding
- Convergence and convex functions

Page Summary

YouTube video player

Machine Learning Crash Course: Gradient Descent

Google for Developers

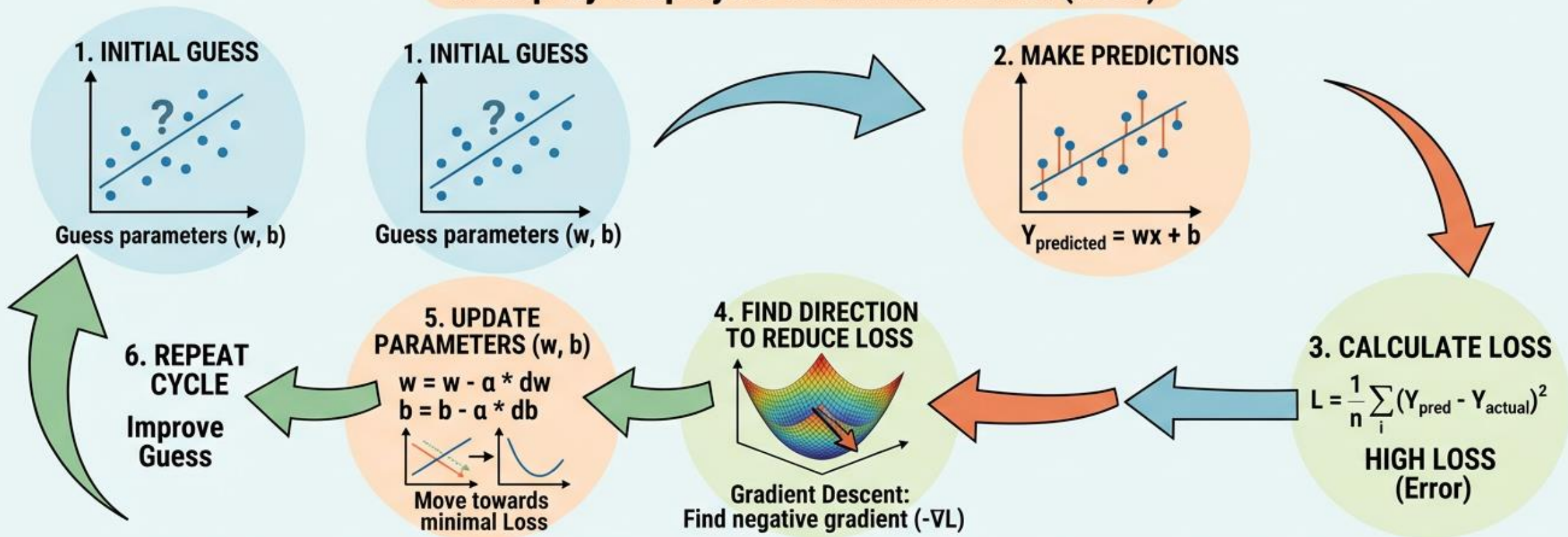
GRADIENT DESCENT

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Complete the GD 10 mins reading

LINEAR REGRESSION: AN ITERATIVE TRAINING PROCESS

A step-by-step cycle to minimize Error (Loss)



KEY TAKEAWAYS



Goal: Minimize Loss



Gradient shows direction



Update parameters & b

Situation	Choose	Why?
Small dataset	OLS	Fast and direct
Need exact OLS coefficients	OLS	Gives the analytical least-squares solution
Very large dataset	Gradient Descent	OLS matrix calculations can become expensive
Dataset doesn't fit comfortably in memory	Gradient Descent	Can use mini-batches
Want to use the same optimization idea for many ML models	Gradient Descent	General-purpose optimizer
Neural networks	Gradient Descent / variants	OLS isn't applicable

Numerical of Linear Regression

Problem 1

Question: Find linear regression equation for the following two sets of data:

x	2	4	6	8
y	3	7	5	10

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x	2	4	6	8
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Linear regression is given by:

$$y = a + bx$$

$$y = 1.5 + 0.95 x$$

Numerical of Linear Regression

Problem 2

Student	S1	S2	S3	S4	S5	S6	S7	S8
First Test, x	56	75	61	61	67	72	62	61
Second Test, y	21	39	34	21	32	24	29	24

$$\sum x = 515, \sum y = 224, \sum x^2 = 33441, \sum y^2 = 6576 \text{ and } \sum xy = 14590.$$

Numerical of Linear Regression

Problem 2

Student	S1	S2	S3	S4	S5	S6	S7	S8
First Test, x	56	75	61	61	67	72	62	61
Second Test, y	21	39	34	21	32	24	29	24

$$\sum x = 515, \sum y = 224, \sum x^2 = 33441, \sum y^2 = 6576 \text{ and } \sum xy = 14590.$$

$$\hat{y} = -10.106 + 0.590x.$$

Numerical of Linear Regression Problem 3

<https://serc.carleton.edu/mathyouneed/geomajors/regression/practice.html>

Math You Need - Majors. (2023, August 11). *Linear Regression: Practice Problems*. <https://serc.carleton.edu/mathyouneed/geomajors/regression/practice.html>

Cost/Error Function

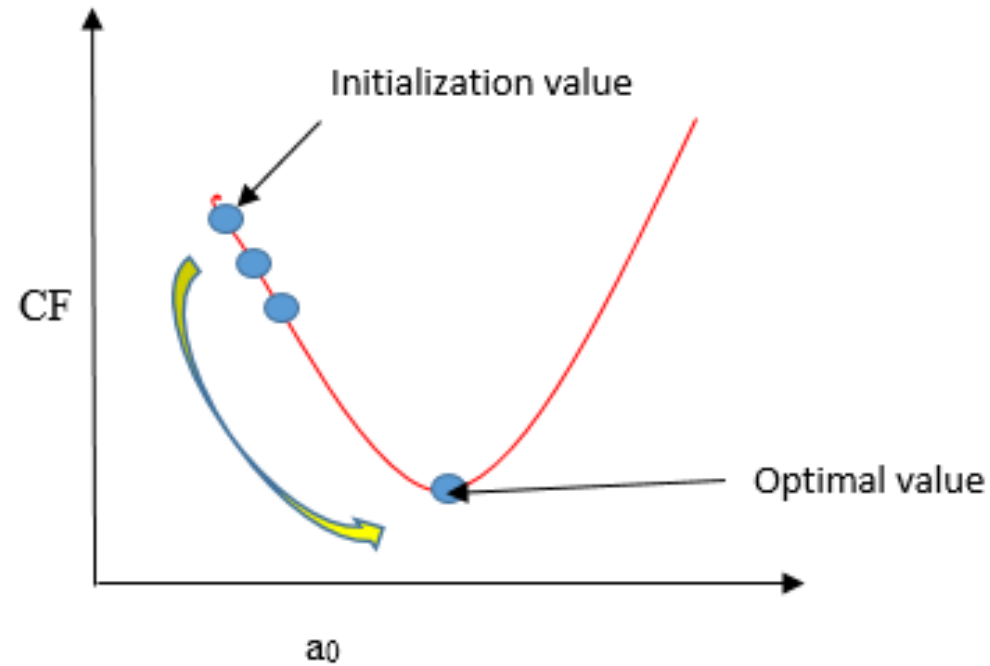
- The cost function helps to figure out the best possible values for a_0 and a_1 , which provides the best fit line for the data points.
- Cost function optimizes the regression coefficients or weights and measures how a linear regression model is performing. The cost function is used to find the accuracy of the **mapping function** that maps the input variable to the output variable. This mapping function is also known as **the Hypothesis function**.
- In Linear Regression, Mean Squared Error (MSE) cost function is used, which is the average of squared error that occurred between the predicted values and actual values.

Let's: y_i = actual values and $mx_i + b$ = predicted value

$$MSE = \frac{1}{N} \sum_{i=1}^n (y_i - (mx_i + b))^2$$

Gradient Descent

- Gradient descent is a method of updating a_0 and a_1 to minimize the cost function (MSE).
- A regression model uses gradient descent to update the coefficients of the line (a_0 , a_1 $\Rightarrow$ x_i , b) by reducing the cost function by a random selection of coefficient values and then iteratively update the values to reach the minimum cost function.



Gradient Descent

- Imagine a pit in the shape of U. You are standing at the topmost point in the pit, and your objective is to reach the bottom of the pit. There is a treasure, and you can only take a discrete number of steps to reach the bottom.
- If you decide to take one footstep at a time, you will eventually get to the bottom of the pit but, this will take a longer time.
- If you choose to take longer steps each time, you may get to sooner but, there is a chance that you could overshoot the bottom of the pit and not near the bottom.
- In the gradient descent algorithm, the number of steps you take is the learning rate, and this decides how fast the algorithm converges to the minima.

Gradient Descent

To update a_0 and a_1 , we take gradients from the cost function. To find these gradients, we take partial derivatives for a_0 and a_1

$$J = \frac{1}{n} \sum_{i=1}^n (a_0 + a_1 \cdot x_i - y_i)^2$$

$$\frac{\partial J}{\partial a_0} = \frac{2}{n} \sum_{i=1}^n (a_0 + a_1 \cdot x_i - y_i)$$

$$\frac{\partial J}{\partial a_1} = \frac{2}{n} \sum_{i=1}^n (a_0 + a_1 \cdot x_i - y_i) \cdot x_i$$

$$\frac{\partial J}{\partial a_0} = \frac{2}{n} \sum_{i=1}^n (pred_i - y_i)$$

$$\frac{\partial J}{\partial a_1} = \frac{2}{n} \sum_{i=1}^n (pred_i - y_i) \cdot x_i$$

$$a_0 = a_0 - \alpha \cdot \frac{2}{n} \sum_{i=1}^n (pred_i - y_i)$$

$$a_1 = a_1 - \alpha \cdot \frac{2}{n} \sum_{i=1}^n (pred_i - y_i) \cdot x_i$$

partial derivatives are the gradients and they are used to update the

The partial derivatives are the gradients, and they are used to update the values of a_0 and a_1 .

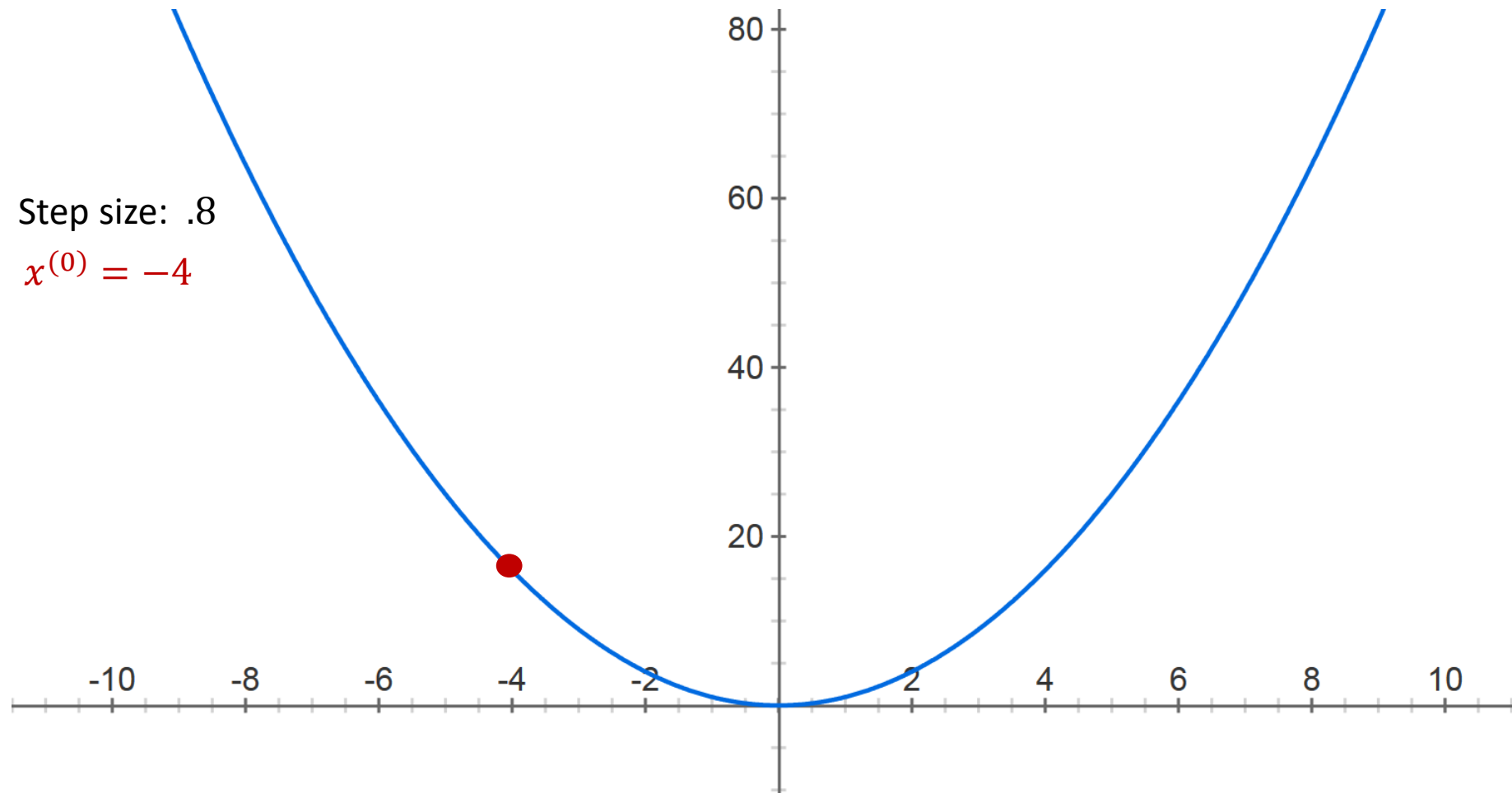
Alpha is the learning rate.

Gradient Descent

$$f(x) = x^2$$

Step size: .8

$$x^{(0)} = -4$$



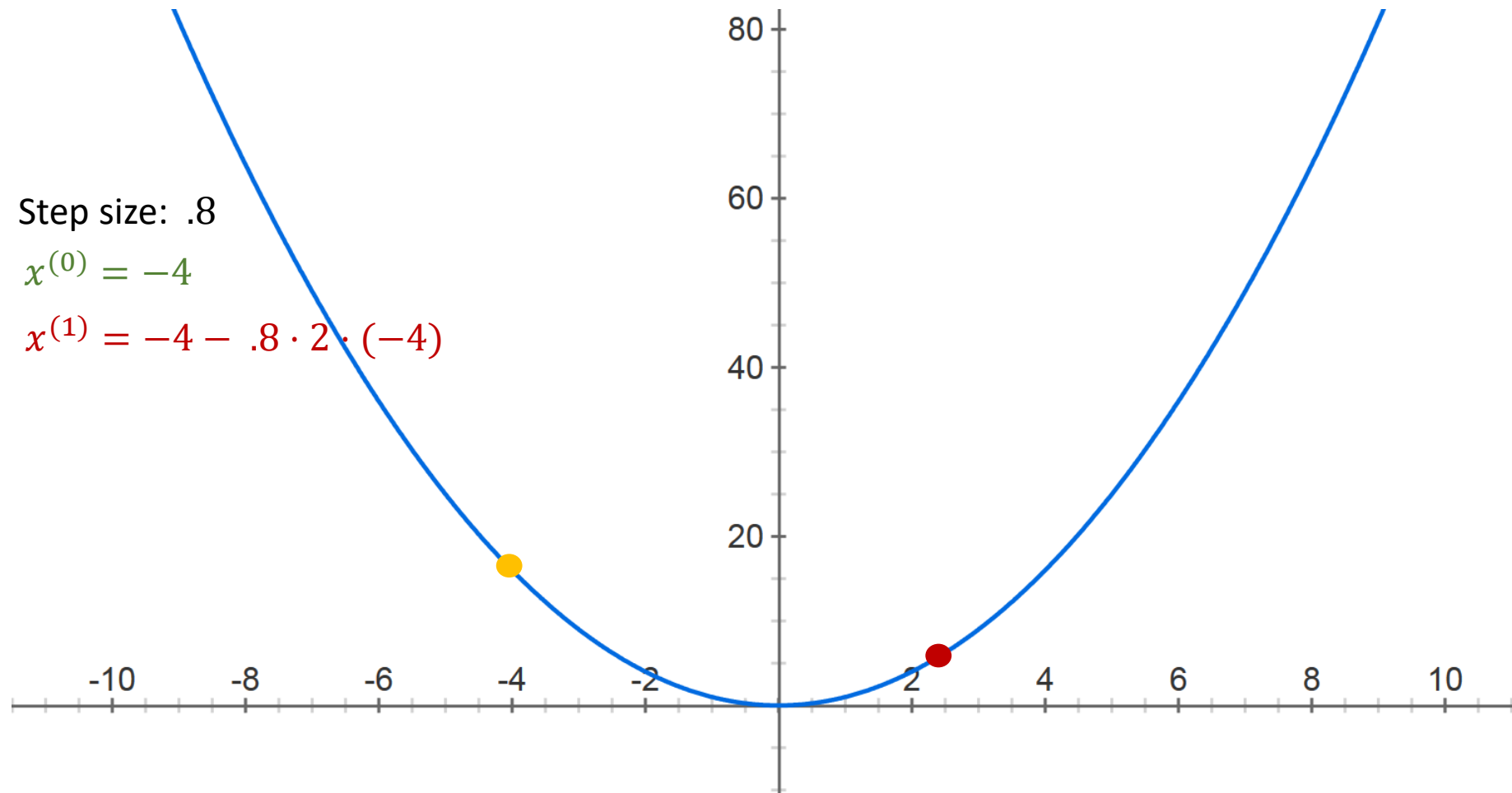
Gradient Descent

$$f(x) = x^2$$

Step size: .8

$$x^{(0)} = -4$$

$$x^{(1)} = -4 - .8 \cdot 2 \cdot (-4)$$



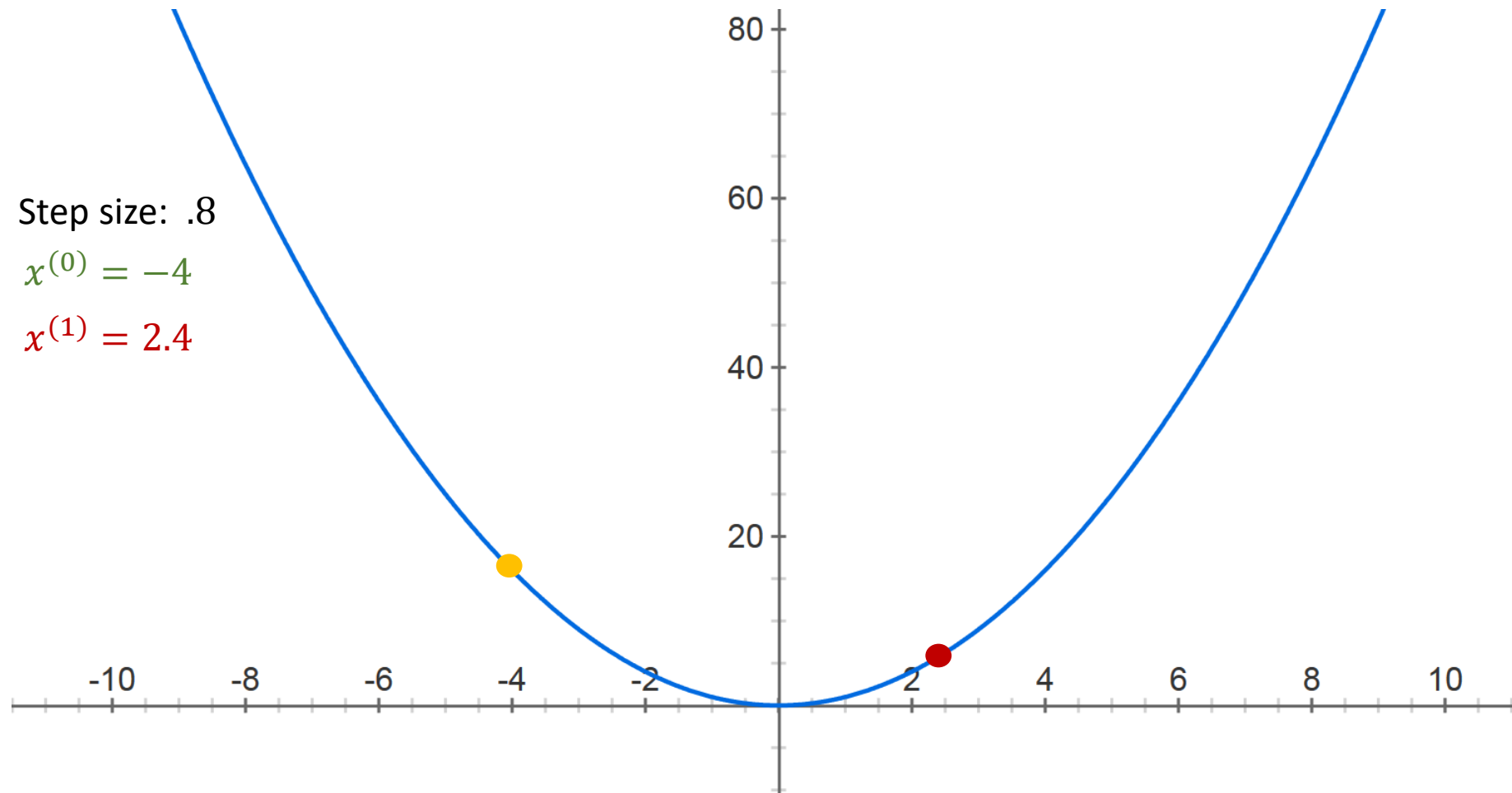
Gradient Descent

$$f(x) = x^2$$

Step size: .8

$$x^{(0)} = -4$$

$$x^{(1)} = 2.4$$



Gradient Descent

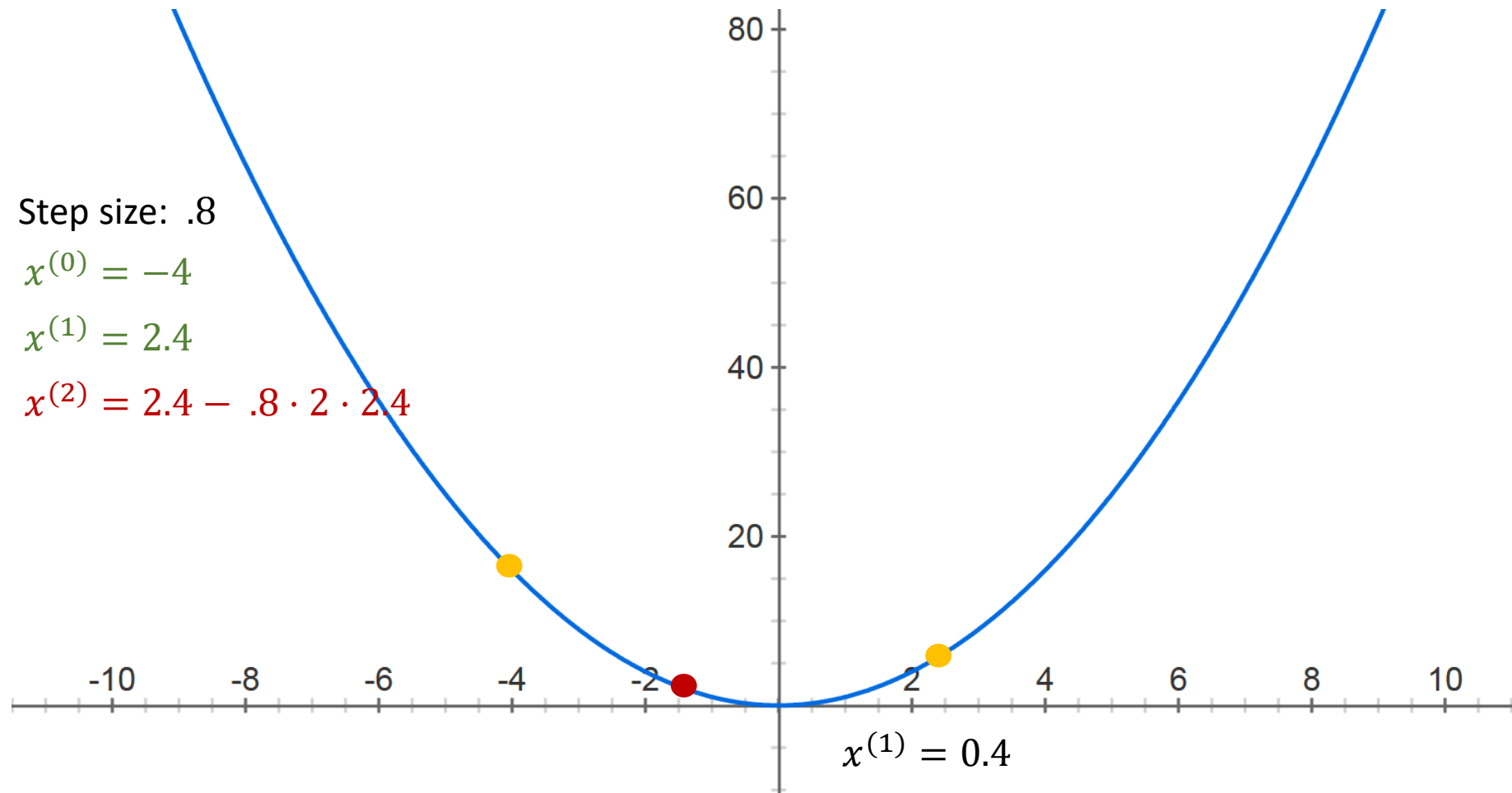
$$f(x) = x^2$$

Step size: .8

$$x^{(0)} = -4$$

$$x^{(1)} = 2.4$$

$$x^{(2)} = 2.4 - .8 \cdot 2 \cdot 2.4$$



Gradient Descent

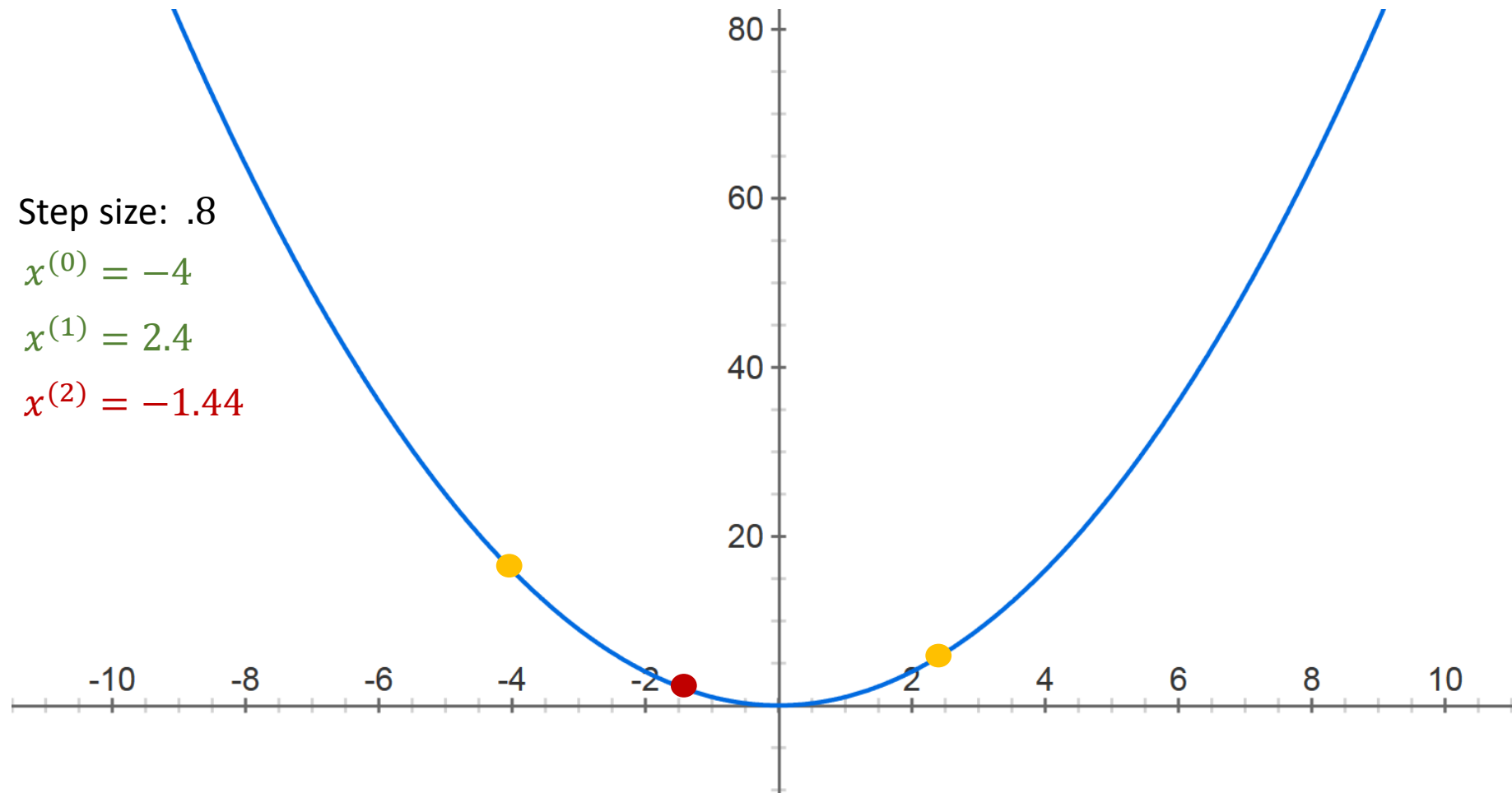
$$f(x) = x^2$$

Step size: .8

$$x^{(0)} = -4$$

$$x^{(1)} = 2.4$$

$$x^{(2)} = -1.44$$



Gradient Descent

$$f(x) = x^2$$

Step size: .8

$$x^{(0)} = -4$$

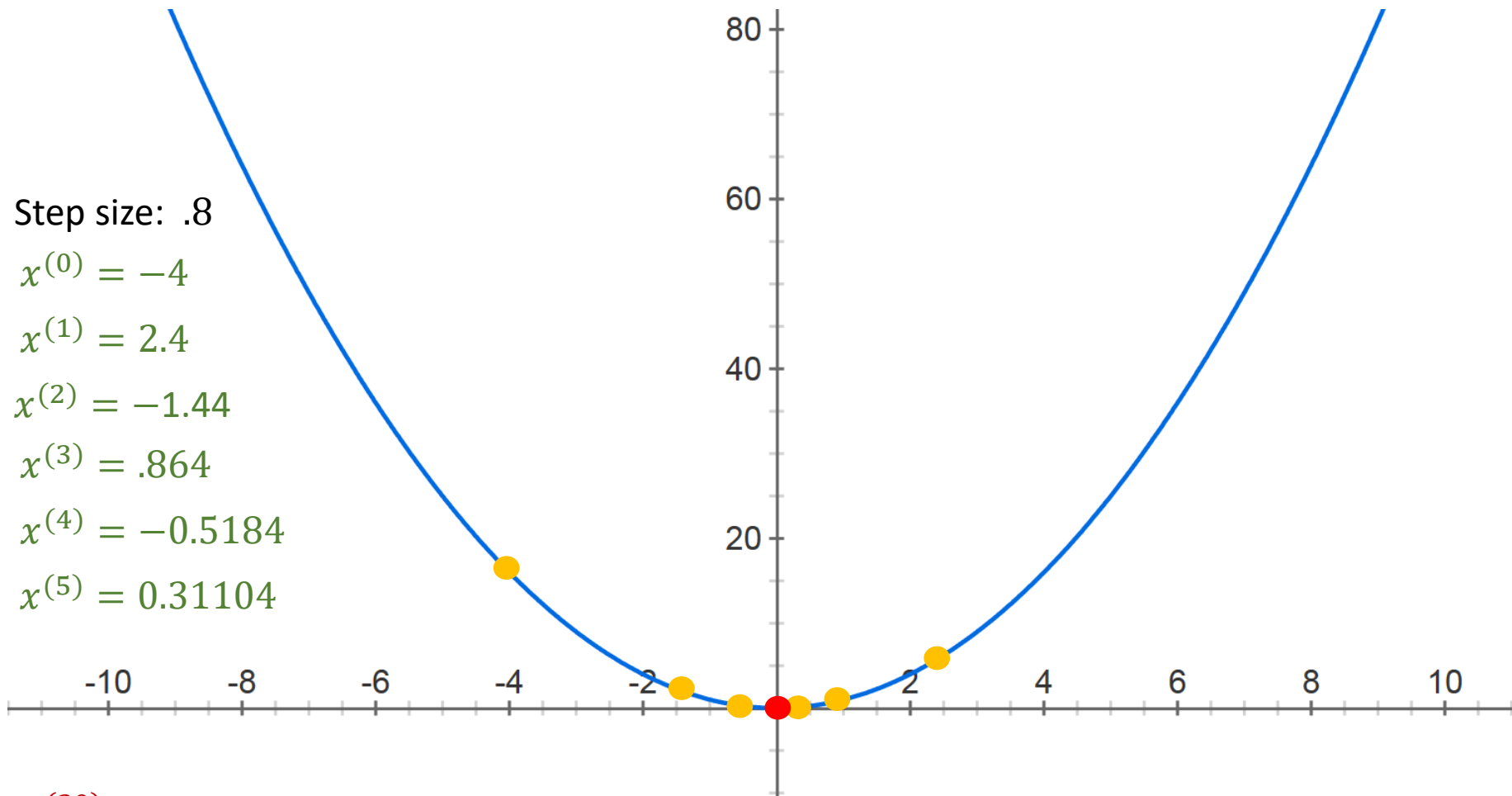
$$x^{(1)} = 2.4$$

$$x^{(2)} = -1.44$$

$$x^{(3)} = .864$$

$$x^{(4)} = -0.5184$$

$$x^{(5)} = 0.31104$$



$$x^{(30)} = -8.84296e - 07$$



numiqo

Linear

Non Linear



Assumptions of

Linear Regression



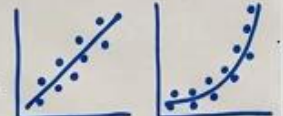
Assumptions of LR

1. List the assumptions of linear regression.
2. Draw an example of a linear and non-linear dataset(X, y).
- 3.a. What can the normal distribution of the error be if the dataset is to be considered for linear regression?
- 3.b. Which plot can help you identify it?
- 4.a. Define homoscedasticity and its types.
- 4.b. Which plot can help you to identify it?
- 5.a. Define Multicollinearity.
- 5.b. Which factor are calculated to check the multicollinearity.
- 5.c. What is the range of VIF to check?

Assumptions of LR

Linearity, Normality (of residuals),
Homoscedasticity, Independence
(of errors), No multicollinearity

1. List the assumptions of linear regression.

2. Draw an example of a linear and non-linear dataset (X,y). 

3.a. What can the normal distribution of the error be if the dataset is to be considered for linear regression? → Residuals should be normally distributed with mean ~ 0 .

Histogram of residuals, Q-Q Plot.

3.b. Which plot can help you identify it?

4.a. Define homoscedasticity and its types.

Homoscedasticity = Constant variance of residuals

Types: Violation is Heteroscedasticity.

4.b. Which plot can help you to identify it?

Residuals vs. Fitted plot, Spread-Level plot.

5.a. Define Multicollinearity. Two or more independent variables are highly correlated.

5.b. Which factor are calculated to check the multicollinearity.

VIF (Variance Inflation Factor), Tolerance,
Correlation matrix.

5.c. What is the range of VIF to check?

VIF > 5 (or sometimes 10) indicates high multicollinearity.

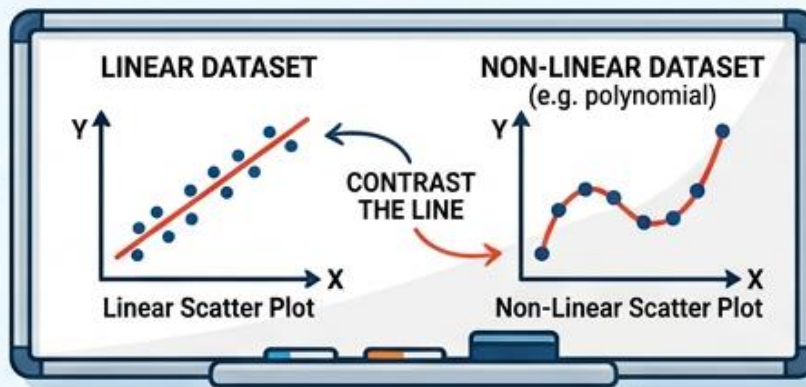


ASSUMPTIONS OF LINEAR REGRESSION

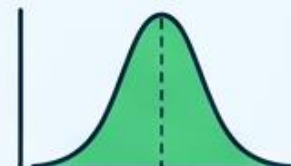
1. LIST THE ASSUMPTIONS

- **LINEAR RELATIONSHIP** (between X & Y)
- **NORMALITY OF ERRORS** (Normal distribution)
- **HOMOSCEDASTICITY** (Constant variance)
- **NO MULTICOLLINEARITY** (Predictors not correlated)
- **INDEPENDENCE OF OBSERVATIONS** (Residuals not correlated)
- **NO ENDOGENEITY** (X and errors uncorrelated).

2. DRAW A LINEAR & NON-LINEAR DATASET



3.a. NORMAL DISTRIBUTION OF ERROR

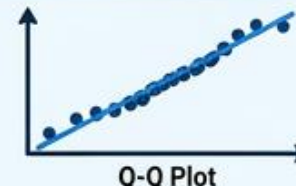


NORMAL ERROR DISTRIBUTION

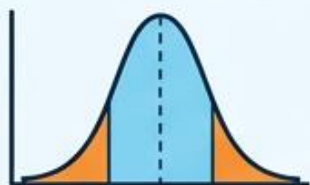
Error Distribution: Should be bell-shaped, symmetrical, centered around zero.

3.b. WHICH PLOT CAN HELP IDENTIFY IT?

Q-Q PLOT



3.a. NORMAL DISTRIBUTION OF ERROR



NORMAL ERROR DISTRIBUTION

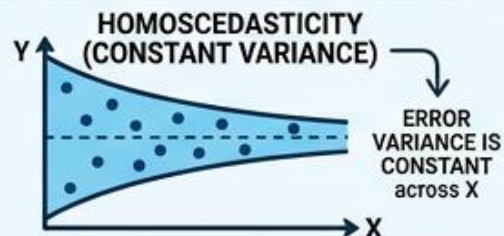
Error Distribution: Should be bell-shaped, symmetrical, centered around zero.

3.b. WHICH PLOT CAN HELP IDENTIFY IT?

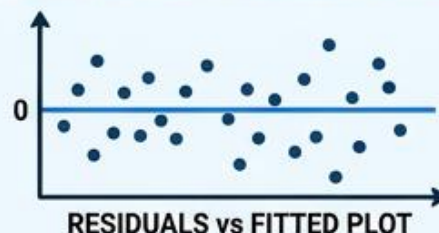
Q-Q PLOT



4.a. DEFINE HOMOSCEDASTICITY



4.b. WHICH PLOT CAN HELP IDENTIFY IT?



5.a. DEFINE MULTICOLLINEARITY



PREDICTOR RELATIONSHIP

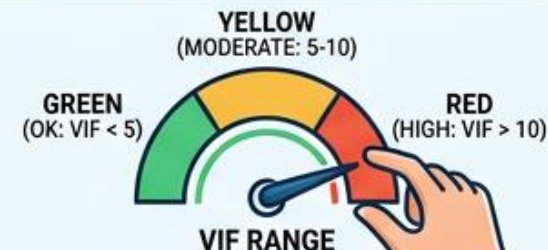
MULTICOLLINEARITY:
High correlation between two or more independent variables (X1, X2, etc.)

5.b. WHICH FACTOR ARE CALCULATED TO CHECK MULTICOLLINEARITY?



VIF (VARIANCE INFLATION FACTOR)

5.c. WHAT IS THE RANGE OF VIF TO CHECK?



Typically, VIF < 10 (or even < 5) is desired to minimize multicollinearity.

To calculate the performance of regression models, we utilize mathematical formulas that will compare the plotted graph to the points that we are predicting.

A good regression model should focus on minimizing the difference between the observation and the predicted value, while being unbiased. (Unbiased means that we try to find a balance between over-estimation and under-estimation)

There are 3 main metrics for model evaluation in regression:

1. R^2 - Coefficient of determination
2. Adjusted R^2
3. Mean Square Error(MSE)/Root Mean Square Error(RMSE)
4. Mean Absolute Error(MAE)

R² Coefficient of Determination /Adjusted R Square

R² measures how much variability in dependent variable can be explained by the model. It is the square of the Correlation Coefficient(R) and that is why it is called R Square.

$$R^2 = 1 - \frac{SS_{Regression}}{SS_{Total}} = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$$

R Square is calculated by the sum of squared of prediction error divided by the total sum of the square which replaces the calculated prediction with mean. R Square value is between 0 to 1 and a bigger value indicates a better fit between prediction and actual value.

Mean Square Error(MSE) / Root Mean Square Error(RMSE)

While R Square is a relative measure of how well the model fits dependent variables, Mean Square Error is an absolute measure of the goodness for the fit.

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

MSE is calculated by the sum of square of prediction error which is real output minus predicted output and then divide by the number of data points. It gives you an absolute number on how much your predicted results deviate from the actual number.

Root Mean Square Error(RMSE) is the square root of MSE.

Mean Absolute Error(MAE)

Mean Absolute Error(MAE) is similar to Mean Square Error(MSE). However, instead of the sum of square of error in MSE, MAE is taking the sum of the absolute value of error.

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|$$

Compared to MSE or RMSE, MAE is a more direct representation of sum of error terms.

MSE gives larger penalization to big prediction error by square it while MAE treats all errors the same.

Mean Absolute Error(MAE)

Mean Absolute Error(MAE) is similar to Mean Square Error(MSE). However, instead of the sum of square of error in MSE, MAE is taking the sum of the absolute value of error.

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|$$

Compared to MSE or RMSE, MAE is a more direct representation of sum of error terms.

MSE gives larger penalization to big prediction error by square it while MAE treats all errors the same.

Metric	Range	Penalizes Large Errors?	Units	Best Value
R^2	0 to 1 (or negative)	✗ No	None (dimensionless)	1
MSE	0 to ∞	✓ Yes	Squared of target variable	Lower is better
RMSE	0 to ∞	✓ Yes (more than MAE)	Same as target variable	Lower is better
MAE	0 to ∞	✗ No	Same as target variable	Lower is better

Metric	Poor Model	Acceptable Model	Good Model	Excellent Model
R^2	< 0.4	$0.4 - 0.6$	$0.6 - 0.8$	> 0.8
RMSE	Close to mean of target variable	~50% of mean	Much lower than mean	Close to 0
MSE	High	Moderate	Low	Very Low
MAE	High	Moderate	Low	Very Low

1. **Dependent Variable:** The main factor in Regression analysis which we want to predict is called the dependent variable. It is also called **target variable**.
2. **Independent Variable:** The factors which affect the dependent variables, or which are used to predict the values of the dependent variables are called independent variable, also called as a **predictor**.
3. **Outliers:** Outlier is an observation which contains either very low value or very high value in comparison to other observed values. An outlier may hamper the result, so it should be avoided.
4. **Multicollinearity:** If the independent variables are highly correlated with each other than other variables, then such condition is called Multicollinearity. It should not be present in the dataset, because it creates problem while ranking the most affecting variable.
5. **Underfitting and Overfitting:** If our algorithm works well with the training dataset but not well with test dataset, then such problem is called **Overfitting**. And if our algorithm does not perform well even with training dataset, then such problem is called **underfitting**.

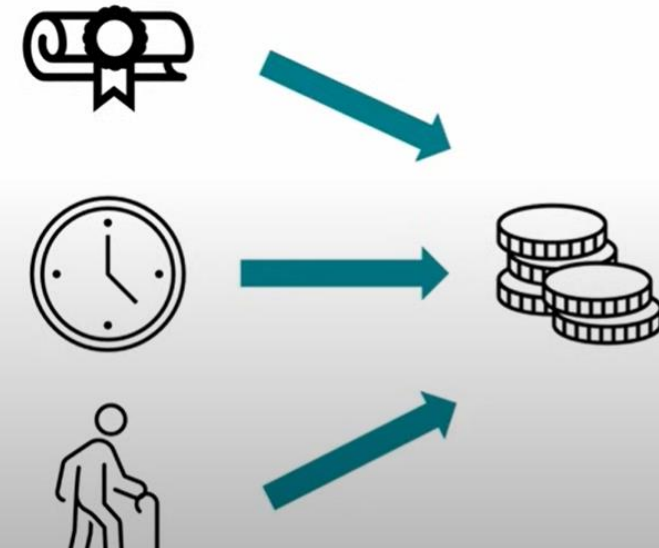
Multiple Linear Regression

Multiple linear regression is used to estimate the relationship between **two or more independent variables** and **one dependent variable**.

Simple linear
Regression



Multiple linear
Regression



Multiple linear regression makes all of the same assumptions as simple linear regression:

- Homogeneity of variance (homoscedasticity): the size of the error in our prediction doesn't change significantly across the values of the independent variable.
- Independence of observations: the observations in the dataset were collected using statistically valid methods, and there are no hidden relationships among variables.
- In multiple linear regression, it is possible that some of the independent variables are actually correlated with one another, so it is important to check these before developing the regression model. If two independent variables are too highly correlated ($r^2 > \sim 0.6$), then only one of them should be used in the regression model.
- Normality: The data follows a normal distribution.
- Linearity: the line of best fit through the data points is a straight line, rather than a curve or some sort of grouping factor.

Multiple Linear Regression Formula

Multiple linear regression formula

$$y = \beta_0 + \beta_1 X_1 + \dots + \beta_n X_n + \varepsilon$$

- **y** = the predicted value of the dependent variable
- **B₀** = the y-intercept (value of y when all other parameters are set to 0)
- **B₁X₁** = the regression coefficient (B₁) of the first independent variable (**X₁**) (a.k.a. the effect that increasing the value of the independent variable has on the predicted **y** value)
- ... = do the same for however many independent variables you are testing
- **B_nX_n** = the regression coefficient of the last independent variable
- **e** = model error (a.k.a. how much variation there is in our estimate of **y**)

Regression Types

There are two main types of regression:

Simple Regression

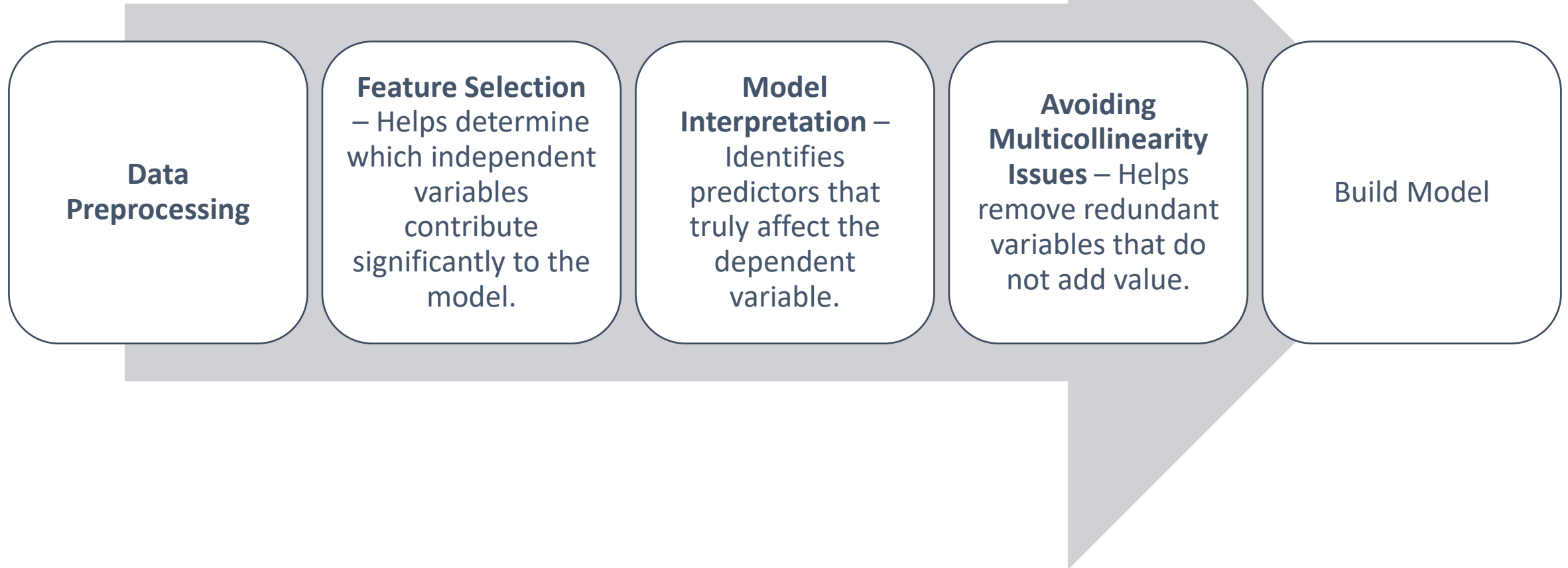
Used to predict a continuous dependent variable based on a single independent variable.
Simple linear regression should be used when there is only a single independent variable.

Multiple Regression

Used to predict a continuous dependent variable based on multiple independent variables.
Multiple linear regression should be used when there are multiple independent variables.

NonLinear Regression

Relationship between the dependent variable and independent variable(s) follows a nonlinear pattern.
Provides flexibility in modeling a wide range of functional forms.



Feature Selection : Why are t-test and p-test Important in Multiple Regression

	P-value	T-value (a.k.a. T-score)
Definition	Probability of getting a result equal to or more extreme than what was observed,	Ratio between means of sample sets and the variation within the sample sets
Formula	$P(\text{Observed Data} \mid \text{Null Hypothesis is true})$	$t \text{ value} = \frac{\overset{\text{sample mean}}{\bar{x}} - \overset{\text{population mean}}{\mu_0}}{\frac{\overset{\text{sample standard deviation}}{s}}{\overset{\text{sample size}}{\sqrt{n}}}}$
Value Range	[0,1]	[-inf, inf]
Interpretation	The smaller the p-value, the less likely that	The larger the t-value, the more different are the groups
Note to self	The smaller the better	The bigger the better

- **Primary criterion:** Use **p-value** for filtering out non-significant features.
- **Secondary criterion:** Use **t-value** to rank and prioritize important features.

✓ Use p-value for feature selection

- If **p-value** < **0.05**, the feature is statistically significant and should be **kept** in the model.
- If **p-value** > **0.05**, the feature is **not significant** and can be **removed**.

Feature	OLS (Ordinary Least Squares)	Lasso (L1 Regularization)	Ridge (L2 Regularization)	Component Regression (PCR)
Objective	Minimizes sum of squared residuals	Minimizes residuals + absolute penalty ($L1$)	Minimizes residuals + squared penalty ($L2$)	Reduces dimensionality using PCA before regression
Feature Selection	No selection, keeps all variables	Shrinks some coefficients to zero (eliminates irrelevant variables)	Shrinks coefficients but does not make them zero	Uses transformed principal components instead of original features
Handles Multicollinearity?	No, suffers from high multicollinearity	Yes, removes redundant features	Yes, reduces impact of correlated variables	Yes, transforms correlated variables into independent components
Overfitting Risk	High (especially with many features)	Low (performs feature selection)	Medium (reduces overfitting but retains all variables)	Low (removes noise but loses some interpretability)
Interpretability	High (keeps original variables)	Medium (some variables removed) ↓	High (keeps all variables)	Low (new variables are transformed components)

Multiple Linear Regression Methods for Coefficient Calculations

1. Machine Learning for Dummies, By John Paul Mueller and Luca Massaron, For Dummies, 2016.

Thank You